

Smart Waste Detection and Segregation Platform

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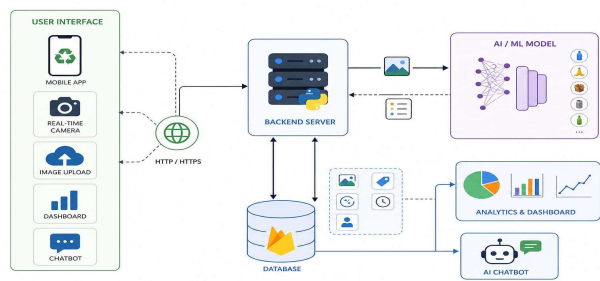
Problem Statement

Rapid urbanization has led to a significant rise in municipal waste, while improper segregation at the source reduces recycling efficiency and increase environmental pollution. This creates a strong need for an intelligent, automated system that can ensure efficient waste classification and support sustainable waste management practices.

Objectives

- Promote sustainable waste management through reduced manual effort, improved efficacy of segregation and increased community involvement via data-driven insights and awareness-based features.
- Create a fully AI-enabled system to detect and classify the various categories of waste with a high level of accuracy using computer vision techniques
- Design / develop a waste detection model that is integrated with a user-friendly web-based platform enabling users to upload and/or take photos of waste items for classification and disposal instruction

Methodology

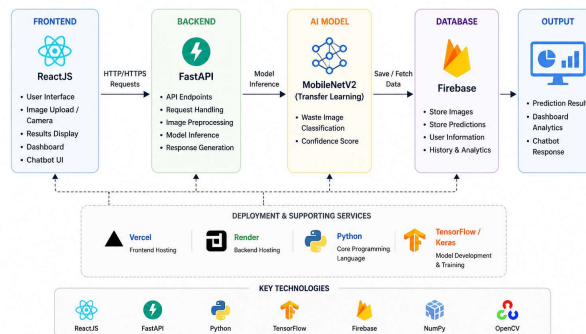


The system uses a MobileNetV2 model with transfer learning to classify waste images. Input images are preprocessed and passed to the model, which predicts the waste category with a confidence score

Implementation

The system is built using ReactJs for the frontend and FastAPI for the backend, with the MobileNetV2 model implemented using TensorFlow/Keras. Key components include image input and preprocessing, model inference for classification, and modules for result display, dashboard analytics, and chatbot interaction

IMPLEMENTATION - BLOCK DIAGRAM



Results

The system achieves ~94% accuracy across 10 waste categories with real-time predictions and confidence scores. Outputs include waste classification, disposal suggestions, and dashboard insights, supported by performance metrics and visual analysis

CONFUSION MATRIX:

[	93	0	0	0	0	1	0	0	0	0]
[	0	92	0	1	3	0	1	0	2	0]
[	1	1	167	1	1	0	10	1	0	0]
[	0	0	0	530	0	0	0	0	2	0]
[	0	0	0	0	285	9	0	11	0	1]
[	0	0	1	0	2	94	1	3	0	1]
[	2	0	6	2	0	0	156	0	0	1]
[	1	1	3	1	7	1	3	178	0	3]
[	0	1	0	2	0	2	0	0	192	0]
[	0	0	1	1	0	3	8	5	2	74]

Confusion Matrix

CLASSIFICATION REPORT:

	precision	recall	f1-score	support
battery	0.96	0.99	0.97	94
biological	0.97	0.93	0.95	99
cardboard	0.94	0.92	0.93	182
clothes	0.99	1.00	0.99	532
glass	0.96	0.93	0.94	306
metal	0.85	0.92	0.88	102
paper	0.87	0.93	0.90	168
plastic	0.90	0.90	0.90	198
shoes	0.97	0.97	0.97	197
trash	0.93	0.79	0.85	94
accuracy				1972
macro avg	0.93	0.93	0.93	1972
weighted avg	0.94	0.94	0.94	1972

Classification Report

Conclusion & Future Scope

Conclusion :

- Developed an AI-based waste classification system using MobileNetV2 with high accuracy and real-time performance.
- Successfully achieved project objectives including automated detection, user-friendly interface, and system integration.
- Reduces manual effort and provides efficient waste management with useful insights and suggestion

Future Scope:

- Integrate IOT-based smart bins and enable real-time video detection for enhanced automation.
- Expand to more waste categories and further improve model accuracy and performance
- Deploy on scalable cloud infrastructure and integrate with smart city system for real world implementation

References

- Bingbing Fang et al. , *Artificial Intelligence for Waste Management in Smart Cities: A Review*, 2023.
- Y. Li et al., *Enhancing Waste Sorting & Recycling Efficiency using Deep Learning Based Approaches*, 2024
- Ahmed at al., *Iot-Based Intelligence Waste Management System*, 2023
- TensorFlow Documentation , MobileNetV2 Model Architecture
- Keras application, Deep learning Models for image classification

Publications

- [1] T. Utreja, M. Vaidya, M. Sharma, and A. kumar, "Smart Waste Detection and Segregation Platform Using Deep Learning and Web-Based Deployment," Communicated to the Project Supervisor, 2026. Status : **Communicated**